

API BDD Test Scenarios: RetryPlaywright Ecommerce Project

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Project: RetryPlaywright

Automation file: tests/APIProjectScenarios.spec.ts

API base URL: https://www.rahulshettyacademy.com/api/ecom

Purpose

This document expresses the project API test coverage as BDD scenarios. The scenarios cover ecommerce API authentication, order creation, customer order retrieval, and authorization behavior.

Feature File

Feature: Ecommerce API validations

As an ecommerce platform consumer

I want the API to authenticate users and manage orders correctly

So that customer order workflows are secure and reliable

Background:

Given the ecommerce API base URL is "https://www.rahulshettyacademy.com/api/ecom"

@API @auth @positive @smoke

Scenario: Registered user logs in successfully through the API

Given a registered API user exists with email "martin.ampah@ashesi.edu.gh"

And the registered API user password is "French123"

When the client sends a POST request to "/auth/login" with the registered credentials

Then the API response should be successful

And the response body should contain an authentication token

And the response body should contain a user ID

And the response message should contain "Login Successfully"

And the token should have three dot-separated segments

@API @auth @negative

Scenario: Unregistered user login is rejected through the API

Given an unregistered API user email is "unregistered.user@example.com"

And the API login password is "French123"

When the client sends a POST request to "/auth/login" with the unregistered credentials

Then the API response should not be successful

And the response status should be 400 or higher

And the response body should contain an error message

And the response message should contain "Incorrect email or password"

@API @orders @positive

Scenario: Authenticated user creates an order through the API

Given the client has logged in through the API with valid credentials

And the client has a valid authentication token

And the order payload contains country "Ghana"

And the order payload contains product ordered ID "6960eaelc941646b7a8b3ed3"

When the client sends a POST request to "/order/create-order" with the order payload

Then the API response should be successful

And the response message should contain "Order Placed Successfully"

And the response body should contain exactly one order ID

And the returned order ID should not be empty

@API @orders @positive

Scenario: Authenticated user retrieves customer orders through the API

Given the client has logged in through the API with valid credentials

And the client has a valid authentication token

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And the client has the authenticated user ID
When the client sends a GET request to "/order/get-orders-for-customer/{userId}"
Then the API response should be successful
And the response body should contain a "data" property
And the "data" property should be an array

@API @orders @security @negative
Scenario: Order creation is rejected when authorization token is missing
  Given the client does not have an authentication token
  And the order payload contains country "Ghana"
  And the order payload contains product ordered ID "6960eae1c941646b7a8b3ed3"
  When the client sends a POST request to "/order/create-order" with the order payload
  Then the API response should not be successful
  And the response status should be 400 or higher
  And the response body should contain an error message
  And no order should be created

```

Traceability Matrix

Scenario	Automated Test Name	File
Registered user logs in successfully through the API	should login registered user and return auth token	tests/APIProjectScenarios.spec.ts
Unregistered user login is rejected through the API	should reject login for unregistered user	tests/APIProjectScenarios.spec.ts
Authenticated user creates an order through the API	should create an order with a valid token	tests/APIProjectScenarios.spec.ts
Authenticated user retrieves customer orders through the API	should retrieve customer orders with a valid token	tests/APIProjectScenarios.spec.ts
Order creation is rejected when authorization token is missing	should reject order creation without authorization token	tests/APIProjectScenarios.spec.ts

Execution Command

```
npx playwright test tests/APIProjectScenarios.spec.ts --project=chromium
```

Current Execution Result

Result	Count
Passed	5
Failed	0

Future BDD API Scenarios To Add

- Scenario: Order creation is rejected when product ordered ID is invalid.
- Scenario: Order creation is rejected when request body is malformed.
- Scenario: Order details are rejected when a user requests another customer's order.
- Scenario: API rejects requests with an invalid token.
- Scenario: API rejects requests with an expired token.